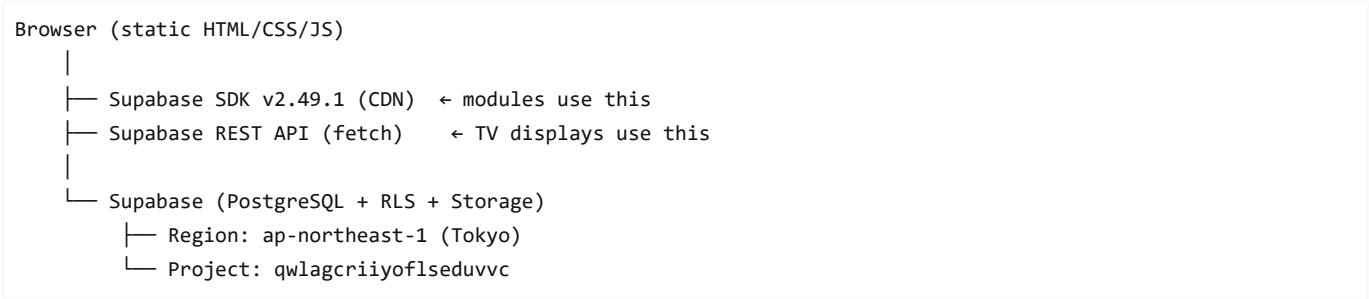


TG FarmHub — Website Structure

Generated 2026-04-10. Reference doc for planning changes.

1. High-Level Architecture



- **No build step** — every page is a self-contained HTML file with inline JS/CSS
- **No framework** — vanilla JS, DOM manipulation, no React/Vue/etc.
- **Hosting:** Netlify at `tgfarmhub.com` (static deploy, `netlify deploy --prod --dir=.`)
- **Auth:** PIN-based for workers/supervisors, Google OAuth for admin. Sessions in localStorage.

2. File Map

Core Application Files

File	Lines	Size	Company	Purpose
index.html	2,119	114 KB	Shared	Hub — login, user management, farm config, module launcher, TV display tab
sales.html	9,645	492 KB	Agro Fruits	Sales — orders, customers, products, payments, invoicing, returns, 7 reports
workers.html	4,567	252 KB	Agribusiness	Workers — profiles, payroll, tasks, loans, expenses, changelog
inventory.html	3,855	204 KB	Agribusiness	Inventory — stock in/out, products, suppliers, stock checks, reports
spraytracker.html	3,647	205 KB	Agribusiness	Spray Tracker — spray jobs, products, AI combos, watchlist summary, reports
seedlings.html	2,746	135 KB	Agribusiness	Seedlings — batches, bookings, collections, L3.1 tracking, MPOB reports
growthtracker.html	582	29 KB	Agribusiness	Growth Tracker — read-only dashboard for block growth data
delivery.html	749	32 KB	Agro Fruits	Delivery — mobile driver page, mark delivered, print DO/CS

TV Displays (standalone, no shared.js)

File	Lines	Size	Purpose
display-sales.html	898	33 KB	Packing station display — 4-column order grid, auto-rotate
display-spray.html	862	34 KB	Spray monitoring display — per-AI pages, auto-rotate

Both TV displays use **Supabase REST API directly** (fetch), not the SDK. Password-gated (sessionStorage).

Shared Infrastructure

File	Lines	Size	Purpose
shared.js	585	21 KB	Supabase init, session guard, sidebar, sbQuery(), sbMutate(), sbUpdateWithLock(), notifications, modals, confirmAction(), calendar picker, offline detection
shared.css	743	19 KB	Sidebar, layout, variables, modal/notification styles, offline banner, typography

Per-Module CSS (extracted from inline)

File	Lines	Purpose
index.css	293	Hub page — login form, module cards, farm config, TV display grid, user management
sales.css	1,331	Sales — cards, badges, timeline, aging, invoicing, A4 documents, receipt
inventory.css	327	Inventory — tables, supplier cards, stock check
seedlings.css	224	Seedlings — batch cards, booking slip, collection forms
spraytracker.css	98	Spray — job cards, summary watchlist
growthtracker.css	83	Growth — dashboard cards, table badges
workers.css	58	Workers — payslip, payroll table
trial-theme.css	473	KIV light theme override (used by sales-trial.html only)

Guides & Docs

File	Purpose
guide-sales.html	Staff workflow guide (English)
guide-sales-cn.html	Staff workflow guide (中文, interface labels in English)
walkthrough-sales.html	Interactive 14-step walkthrough with mockup screens
sales-trial.html	Sales module with trial light theme (KIV)
icon-preview.html	Module icon set preview page

Static Assets

```
assets/
├─ logo.png           (828 KB) – TG Agro Fruits (pineapple)
├─ logo.jpg           (460 KB) – same, JPEG
├─ logo_original.png  (56 KB) – original smaller logo
├─ logo-agribusiness.png (279 KB) – TG Agribusiness (leaf, transparent bg)
├─ logo-agribusiness.jpg (59 KB) – same, cropped JPEG
└─ logo-agribusiness-original.jpg (148 KB) – original with excess whitespace

icons/modules/
├─ farm-config.png    – 3D clay cog with pineapple
├─ growth-tracker.png – 3D clay pineapple plant
```

└─ inventory.png	– 3D clay crate with fertilizer
└─ sales.png	– 3D clay basket of pineapples
└─ seedlings.png	– 3D clay sapling in polybag
└─ spray-tracker.png	– 3D clay spray bottle
└─ tv-display.png	– 3D clay retro TV
└─ workers.png	– 3D clay farmer with pineapple

Planning & Design Docs

docs/superpowers/	
└─ specs/	– Design specs (sales, invoicing, multi-company, packing display, seedlings)
└─ sections/	– 14 invoicing section plans
└─ implementation/	– Code reviews per section
└─ plans/	– Implementation plans (sales phases, multi-company, packing, pcs-order)

3. Module → Database Table Map

Hub (index.html) — Farm Config + User Management

Table	Access
users	R/W — login, user CRUD, permissions
crops	R/W — crop types
crop_varieties	R/W — variety config (days to induce, harvest days)
crop_statuses	R — status list
pnd_blocks	R/W — physical blocks
block_crops	R/W — block-crop assignments, lifecycle, planting cycles
growth_records	R/W — target dates (auto-created on block config)
pnd_blocks	R — for spray summary
pnd_jobs	R — active job counts
pnd_spray_logs	R — spray history
pnd_ingredients	R — for TV display config modal
pnd_tv_config	R/W — watched Als for spray TV
sales_orders	R — order counts for company overview
workers	R — worker counts for company overview
companies	R — company list

Sales (sales.html)

Table	Access
sales_customers	R/W
sales_customer_branches	R/W

sales_products	R/W
sales_orders	R/W
sales_order_items	R/W
sales_payments	R/W
sales_returns	R/W
sales_drivers	R/W
sales_invoices	R/W
sales_invoice_items	R/W
sales_invoice_orders	R/W
sales_invoice_payments	R/W
sales_credit_notes	R/W
crop_varieties	R — for product variety links
workers	R — driver/assignment dropdowns (cross-company)
users	R — for logged-by names

Workers (workers.html)

Table	Access
workers	R/W
worker_roles	R/W
worker_loans	R/W
loan_repayments	R/W
worker_default_responsibilities	R/W
responsibility_types	R/W
payroll_periods	R/W
payroll_entries	R/W
payroll_responsibilities	R/W
salary_advances	R/W
task_entries	R/W
task_types	R/W
task_units	R
employment_stints	R/W
audit_log	R/W
users	R

Inventory (inventory.html)

Table	Access
products	R/W — inventory products
pnd_products	R/W — spray product link
pnd_formulations	R
pnd_ingredients	R
pnd_product_ingredients	R

Spray Tracker (spraytracker.html)

Table	Access
pnd_products	R/W
pnd_product_ingredients	R/W
pnd_jobs	R/W
pnd_spray_logs	R/W
ai_combo_defaults	R/W
products	R/W — inventory link
ingredient_inventory_link	R/W
Views: pnd_latest_sprays, pnd_latest_sprays_by_ai	R

Also reads from shared.js: `pnd_blocks` , `block_crops` , `crop_statuses` , `crop_varieties` , `pnd_ingredients` , `pnd_formulations`

Growth Tracker (growthtracker.html)

Table	Access
pnd_blocks	R
block_crops	R
crop_varieties	R
crop_statuses	R
growth_records_view	R — computed view (days_after_induce, days_to_harvest)

Seedlings (seedlings.html)

Table	Access
seedling_suppliers	R/W
seedling_batches	R/W
seedling_batch_events	R/W

seedling_customers	R/W
seedling_bookings	R/W
seedling_payments	R/W
seedling_collections	R/W
sales_customers	R — cross-company customer link

Delivery (delivery.html)

Table	Access
sales_orders	R/W
sales_order_items	R/W
sales_customers	R
sales_products	R
crop_varieties	R
workers	R
users	R

TV Displays (REST API, read-only)

Display	Tables Read
display-sales.html	sales_orders, sales_order_items, sales_customers, sales_products, workers
display-spray.html	pnd_tv_config, pnd_ingredients, pnd_product_ingredients, pnd_products, pnd_latest_sprays_by_ai, pnd_blocks, block_crops, crop_statuses

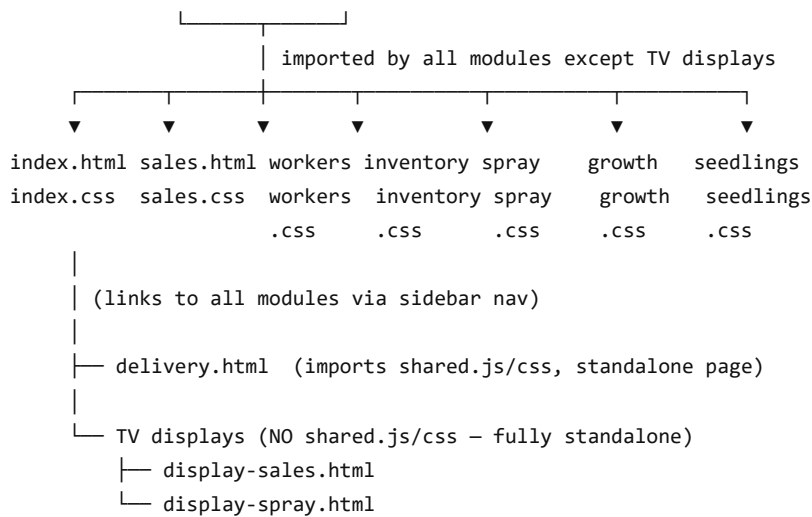
4. Two-Company Structure

Company	Code	ID	Modules
TG Agro Fruits	AF	tg_agro_fruits	Sales, Delivery
TG Agribusiness	AB	tg_agribusiness	Inventory, Workers, Spray Tracker, Growth Tracker, Seedlings

- **Farm Config** is shared (both companies)
- **Hub page** has a company switcher (segmented pill toggle)
- **35 tables** have `company_id` column — all queries filter by it
- **Document numbering:** company-prefixed via `next_id()` RPC (e.g. `AF-S0001` , `AB-W028`)
- **Workers exception:** loaded without company filter in Sales/Delivery (Agro Fruits needs Agribusiness workers for driver dropdowns)

5. Dependency Graph

	shared.js	← Supabase init, session, sidebar, utilities
	shared.css	← Layout, sidebar, variables, modals



External Dependencies (CDN)

- **Supabase JS SDK** `@supabase/supabase-js@2.49.1` — all modules except TV displays
- **html2canvas** — Sales module (document PNG export for WhatsApp sharing)
- **Google Fonts** — Plus Jakarta Sans (shared.css), Inter (TV displays)

6. Supabase Storage Buckets

Bucket	Used By	Content
sales-photos	Sales, Delivery	Prep photos, delivery photos, payment slips (orders/{id}/, payment-slips/)
seedling-photos	Seedlings	Collection photos, L3.1 certificate photos

7. Auth & Session Flow

```

Hub (index.html)
├─ PIN login → lookup users table → store session in localStorage
├─ Google OAuth → Supabase Auth → match email to users table → store session
├─
├─ localStorage keys:
│   ├─ tg_session      - user_id
│   ├─ tg_user         - full user object (name, role, permissions)
│   ├─ tgfarmhub_company - selected company (defaults to tg_agro_fruits)
│   └─ tg_last_user    - last username (for convenience)
├─
└─ Session guard (shared.js):
    ├─ Checks localStorage on every page load
    ├─ Multi-device: session_token polled every 30s
    ├─ Inactivity timeout: 60 minutes
    └─ Redirects to index.html if invalid

TV Displays:
└─ Password gate → sessionStorage (re-prompts on browser restart)
  
```

8. Page Navigation

```
index.html (Hub)
├─ Sidebar modules:
│   └─ sales.html?session={id}
│   └─ inventory.html?session={id}
│   └─ workers.html?session={id}
│   └─ spraytracker.html?session={id}
│   └─ growthtracker.html?session={id}
│   └─ seedlings.html?session={id}
├─ TV Display tab (opens in new tabs):
│   └─ nanasgrowth.tgfarmhub.com (separate repo/deploy)
│   └─ weather.tgfarmhub.com (separate deploy)
│   └─ display-sales.html
│   └─ display-spray.html
├─ delivery.html (linked from Sales, standalone for drivers)
└─ Guides (linked from Sales):
    └─ guide-sales.html
    └─ guide-sales-cn.html
    └─ walkthrough-sales.html
```

9. Theme & Design Tokens

```
--bg:           #FAF6EF /* cream background */
--bg-card:      #FFFFFF /* white cards */
--text:         #2A1A3E /* deep purple/plum */
--green:        #D4AF37 /* ⚠ STALE NAME – actually gold accent */
--green-light:  #6B4C8A /* ⚠ STALE NAME – actually purple secondary */
Font:           'Plus Jakarta Sans' (Google Fonts)
```

- Module icons: 3D clay Pixar-style PNGs (180px source, 84px on cards)
- Company logos: Agro Fruits = pineapple, Agribusiness = leaf
- Sidebar logos are company-specific per module

10. Size Summary

Category	Count	Total Lines	Total Size
Application modules	8	24,910	1.46 MB
TV displays	2	1,760	67 KB
Shared JS/CSS	2	1,328	41 KB
Module CSS	8	2,887	90 KB
Guides/previews	4	1,478	67 KB
Total deployed code	24	32,363	1.73 MB

Largest files by code: `sales.html` (9,645 lines), `sales-trial.html` (4,893), `workers.html` (4,567), `inventory.html` (3,855), `spraytracker.html` (3,647), `seedlings.html` (2,746).

11. Sub-Projects (Separate Repos/Deploys, Same Supabase)

Project	Domain	Repo	Reads From
TG Nanas Growth TV	nanasgrowth.tgfarmhub.com	yapwaylon-sketch/TG-Nanas-Growth-TV	growth_records, pnd_blocks, crop_varieties
TG Weather	weather.tgfarmhub.com	Not on GitHub (deferred)	Weather tables

12. Key Shared Patterns

- `sbQuery()` — wraps all Supabase SDK calls (try-catch + notify + offline check)
- `sbMutate()` — mutation wrapper with retry + exponential backoff
- `sbUpdateWithLock()` — optimistic locking via `updated_at` check
- `confirmAction()` — styled modal replacement for `confirm()` (renders HTML body)
- `dbNextId()` → calls `next_id()` RPC — company-prefixed sequential IDs
- `calOpen()` / `calClose()` — custom calendar date picker
- `getCompanyId()` — reads from `localStorage`, used in every query filter
- `soProcessPhotoFile()` — resize to 1200px, JPEG 80%, upload to Supabase Storage
- All `.select()` chained on mutations (Supabase v2 returns empty without it)